

Example 1

```
supplement_heart_rate |> anovas(outcome = resting_heart_rate_bpm,  
                                A = treatment,  
                                B = pony_type,  
                                interaction = TRUE)
```

Two-way ANOVA (with interaction)

Source	df	SS	MS	F	p-value
treatment	1	215.2813	215.2813	15.409	< 0.001
pony_type	2	1383.6803	691.8401	49.519	< 0.001
treatment:pony_type	2	636.6508	318.3254	22.785	< 0.001
Residuals	66	922.0942	13.9711		

Hypotheses (treatment:pony_type):

H_0 : There is no interaction between treatment and pony_type

H_1 : There is an interaction between treatment and pony_type

Test statistic: $F(2, 66) = 22.785$

p-value: < 0.001

Conclusion: Reject the null hypothesis ($p = < 0.001 < \alpha = 0.050$)

- **Hypotheses:**
- **Test Statistic and p -Value**
- **Rejection Region**
- **Conclusion and interpretation**

Example 2

```
exercise_recovery |> anovas(outcome = recovery_time_min,  
                             A = treatment,  
                             B = pony_type,  
                             interaction = TRUE)
```

Two-way ANOVA (with interaction)

Source	df	SS	MS	F	p-value
treatment	2	2909.4913	1454.7456	79.154	< 0.001
pony_type	2	23.1424	11.5712	0.630	0.535
treatment:pony_type	4	47.5704	11.8926	0.647	0.630
Residuals	99	1819.4858	18.3786		

Hypotheses (treatment:pony_type):

H_0 : There is no interaction between treatment and pony_type

H_1 : There is an interaction between treatment and pony_type

Test statistic: $F(4, 99) = 0.647$

p-value: 0.630

Conclusion: Fail to reject the null hypothesis ($p = 0.630 \geq \alpha = 0.050$)

- **Hypotheses:**
- **Test Statistic and p -Value**
- **Rejection Region**
- **Conclusion and interpretation**

```
exercise_recovery |> anovas(outcome = recovery_time_min,
                             A = treatment,
                             B = pony_type,
                             interaction = FALSE)
```

Two-way ANOVA (no interaction)

Source	df	SS	MS	F	p-value
treatment	2	2909.4913	1454.7456	80.254	< 0.001
pony_type	2	23.1424	11.5712	0.638	0.530
Residuals	103	1867.0562	18.1268		

Hypotheses (treatment):

H_0 : All means across levels of treatment are equal

H_1 : At least one mean across levels of treatment differs

Test statistic: $F(2, 103) = 80.254$

p-value: < 0.001

Conclusion: Reject the null hypothesis ($p = < 0.001 < \alpha = 0.050$)

Hypotheses (pony_type):

H_0 : All means across levels of pony_type are equal

H_1 : At least one mean across levels of pony_type differs

Test statistic: $F(2, 103) = 0.638$

p-value: 0.530

Conclusion: Fail to reject the null hypothesis ($p = 0.530 \geq \alpha = 0.050$)

- **Hypotheses:**

- **Test Statistic and p -Value**

- **Rejection Region**

- **Conclusion and interpretation**

- **Hypotheses:**
- **Test Statistic and p -Value**
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Let's Practice 1

Applejack and the Ponyville clinic are studying whether a high-fiber feed changes postmeal blood glucose. Within each pony type, adult ponies are randomly assigned to Standard feed or High-fiber feed for two weeks. After a standardized meal, glucose is measured once at the same pre-specified time for every pony.

```
glucose |> anovas(outcome = postmeal_glucose_mg_dl,  
                  A = treatment,  
                  B = pony_type)
```

Two-way ANOVA (with interaction)

Source	df	SS	MS	F	p-value
treatment	1	531.4410	531.4410	13.231	< 0.001
pony_type	2	2111.6527	1055.8263	26.286	< 0.001
treatment:pony_type	2	2635.6647	1317.8323	32.809	< 0.001
Residuals	84	3374.0227	40.1669		

Hypotheses (treatment:pony_type):

H_0 : There is no interaction between treatment and pony_type

H_1 : There is an interaction between treatment and pony_type

Test statistic: $F(2, 84) = 32.809$

p-value: < 0.001

Conclusion: Reject the null hypothesis ($p = < 0.001 < \alpha = 0.050$)

```
glucose |> anovas(outcome = postmeal_glucose_mg_dl,  
                  A = treatment,  
                  B = pony_type,  
                  interaction = FALSE)
```

Two-way ANOVA (no interaction)

Source	df	SS	MS	F	p-value
treatment	1	531.4410	531.4410	7.605	0.007
pony_type	2	2111.6527	1055.8263	15.109	< 0.001
Residuals	86	6009.6873	69.8801		

Hypotheses (treatment):

H_0 : All means across levels of treatment are equal

H_1 : At least one mean across levels of treatment differs

Test statistic: $F(1, 86) = 7.605$

p-value: 0.007

Conclusion: Reject the null hypothesis ($p = 0.007 < \alpha = 0.050$)

Hypotheses (pony_type):

H_0 : All means across levels of pony_type are equal

H_1 : At least one mean across levels of pony_type differs

Test statistic: $F(2, 86) = 15.109$

p-value: < 0.001

Conclusion: Reject the null hypothesis ($p = < 0.001 < \alpha = 0.050$)

1. Are you reporting the interaction or main effects? Why?

2. Report the appropriate hypothesis test(s).

Let's Practice 2

Fluttershy is testing a fictional anti-inflammatory treatment in adult ponies with the same mild inflammatory condition. Within each pony type, ponies are randomly assigned to Placebo, Low dose, or High dose. A blood inflammation marker is measured once after seven days.

```
inflammation |> anovas(outcome = inflammation_marker_mg_l,  
                      A = treatment,  
                      B = pony_type)
```

Two-way ANOVA (with interaction)

Source	df	SS	MS	F	p-value
treatment	2	384.4924	192.2462	86.419	< 0.001
pony_type	2	366.1000	183.0500	82.286	< 0.001
treatment:pony_type	4	303.1356	75.7839	34.067	< 0.001
Residuals	126	280.2960	2.2246		

Hypotheses (treatment:pony_type):

H_0 : There is no interaction between treatment and pony_type

H_1 : There is an interaction between treatment and pony_type

Test statistic: $F(4, 126) = 34.067$

p-value: < 0.001

Conclusion: Reject the null hypothesis ($p = < 0.001 < \alpha = 0.050$)

```
inflammation |> anovas(outcome = inflammation_marker_mg_l,  
                      A = treatment,  
                      B = pony_type,  
                      interaction = FALSE)
```

Two-way ANOVA (no interaction)

Source	df	SS	MS	F	p-value
treatment	2	384.4924	192.2462	42.836	< 0.001
pony_type	2	366.1000	183.0500	40.787	< 0.001
Residuals	130	583.4316	4.4879		

Hypotheses (treatment):

H_0 : All means across levels of treatment are equal

H_1 : At least one mean across levels of treatment differs

Test statistic: $F(2, 130) = 42.836$
p-value: < 0.001
Conclusion: Reject the null hypothesis ($p = < 0.001 < \alpha = 0.050$)

Hypotheses (pony_type):
 H_0 : All means across levels of pony_type are equal
 H_1 : At least one mean across levels of pony_type differs
Test statistic: $F(2, 130) = 40.787$
p-value: < 0.001
Conclusion: Reject the null hypothesis ($p = < 0.001 < \alpha = 0.050$)

1. Are you reporting the interaction or main effects? Why?

2. Report the appropriate hypothesis test(s).

Let's Practice 3

Nurse Redheart compares dressings in adult ponies receiving care for comparable minor wounds. Within each pony type, ponies are randomly assigned to Standard dressing, Hydrogel, or Enhanced hydrogel. The clinic records the time from first dressing to complete wound closure using a common clinical criterion. Each pony contributes one wound.

```
wound_healing |> anovas(outcome = wound_closure_days,  
                        A = treatment,  
                        B = pony_type)
```

Two-way ANOVA (with interaction)

Source	df	SS	MS	F	p-value
treatment	2	1306.4801	653.2401	241.137	< 0.001
pony_type	2	944.2704	472.1352	174.284	< 0.001
treatment:pony_type	4	6.6327	1.6582	0.612	0.655
Residuals	126	341.3333	2.7090		

Hypotheses (treatment:pony_type):

H_0 : There is no interaction between treatment and pony_type

H_1 : There is an interaction between treatment and pony_type

Test statistic: $F(4, 126) = 0.612$

p-value: 0.655

Conclusion: Fail to reject the null hypothesis ($p = 0.655 \geq \alpha = 0.050$)

```
wound_healing |> anovas(outcome = wound_closure_days,  
                        A = treatment,  
                        B = pony_type,  
                        interaction = FALSE)
```

Two-way ANOVA (no interaction)

Source	df	SS	MS	F	p-value
treatment	2	1306.4801	653.2401	244.050	< 0.001
pony_type	2	944.2704	472.1352	176.390	< 0.001
Residuals	130	347.9661	2.6767		

Hypotheses (treatment):

H_0 : All means across levels of treatment are equal

H_1 : At least one mean across levels of treatment differs
Test statistic: $F(2, 130) = 244.05$
p-value: < 0.001
Conclusion: Reject the null hypothesis ($p = < 0.001 < \alpha = 0.050$)

Hypotheses (pony_type):
 H_0 : All means across levels of pony_type are equal
 H_1 : At least one mean across levels of pony_type differs
Test statistic: $F(2, 130) = 176.39$
p-value: < 0.001
Conclusion: Reject the null hypothesis ($p = < 0.001 < \alpha = 0.050$)

1. Are you reporting the interaction or main effects? Why?

2. Report the appropriate hypothesis test(s).

Let's Practice 4

Rainbow Dash helps the Ponyville clinic compare two fluid treatments after a standardized exercise session. Within each pony type, adult ponies are randomly assigned to Standard fluids or Electrolyte fluids. Time to reach a prespecified hydration target is recorded once for each pony.

```
rehydration |> anovas(outcome = rehydration_time_min,  
                     A = treatment,  
                     B = pony_type)
```

Two-way ANOVA (with interaction)

Source	df	SS	MS	F	p-value
treatment	1	2301.2890	2301.2890	258.932	< 0.001
pony_type	2	36.5502	18.2751	2.056	0.134
treatment:pony_type	2	2.4427	1.2213	0.137	0.872
Residuals	84	746.5613	8.8876		

Hypotheses (treatment:pony_type):

H_0 : There is no interaction between treatment and pony_type

H_1 : There is an interaction between treatment and pony_type

Test statistic: $F(2, 84) = 0.137$

p-value: 0.872

Conclusion: Fail to reject the null hypothesis ($p = 0.872 \geq \alpha = 0.050$)

```
rehydration |> anovas(outcome = rehydration_time_min,  
                     A = treatment,  
                     B = pony_type,  
                     interaction = FALSE)
```

Two-way ANOVA (no interaction)

Source	df	SS	MS	F	p-value
treatment	1	2301.2890	2301.2890	264.232	< 0.001
pony_type	2	36.5502	18.2751	2.098	0.129
Residuals	86	749.0040	8.7093		

Hypotheses (treatment):

H_0 : All means across levels of treatment are equal

H_1 : At least one mean across levels of treatment differs

Test statistic: $F(1, 86) = 264.232$

p-value: < 0.001

Conclusion: Reject the null hypothesis ($p = < 0.001 < \alpha = 0.050$)

Hypotheses (pony_type):

H_0 : All means across levels of pony_type are equal

H_1 : At least one mean across levels of pony_type differs

Test statistic: $F(2, 86) = 2.098$

p-value: 0.129

Conclusion: Fail to reject the null hypothesis ($p = 0.129 \geq \alpha = 0.050$)

1. Are you reporting the interaction or main effects? Why?

2. Report the appropriate hypothesis test(s).